

Smooth morphisms: generalities, differential properties

SGA 1, Exposé II

English draft

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About this draft

This is an unofficial English translation of Exposé II, “Morphismes lisses: généralités, propriétés différentielles”, from SGA 1. The source volume is A. Grothendieck and M. Raynaud, *Revêtements étales et groupe fondamental* (SGA 1), Séminaire de géométrie algébrique du Bois Marie, 1960–61. It follows the slightly corrected SMF recomposition, arXiv:math/0206203v2.

This draft contains the opening convention, all five sections, and the closing errata, including proofs and footnotes. Statement numbering and mathematical notation follow the source, including its two statements numbered 1.1 (a definition and a proposition) and its two statements numbered 4.18 (a corollary and a block of remarks). References to material outside this draft retain their original numbers. Apparent mathematical misprints in the source have been retained; they are recorded in the accompanying README, separately from the translated text. Scholarly proofreading remains outstanding.

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EXPOSÉ II

Smooth morphisms: generalities, differential properties

Cross-references to Exposé I are indicated by I. One recalls that rings are noetherian, and preschemes locally noetherian.

1. Generalities

Let Y be a prescheme, let $t_1, \dots, t_n$ be indeterminates, one sets

$$(1.1) \quad Y[t_1, \dots, t_n] = Y \otimes_{\mathbf{Z}} \mathbf{Z}[t_1, \dots, t_n].$$

Thus $Y[t_1, \dots, t_n]$ is a Y -scheme, affine over Y , defined by the quasi-coherent sheaf of algebras $\mathcal{O}_Y[t_1, \dots, t_n]$. The datum of a section of this prescheme over Y is therefore equivalent to the datum of n sections of $\mathcal{O}_Y$ (corresponding to the images of the t_i under the corresponding homomorphism). If Y' is over Y , one has

$$(1.2) \quad Y[t_1, \dots, t_n] \times_Y Y' = Y'[t_1, \dots, t_n],$$

(which implies that the datum of a Y -morphism from Y' to $Y[t_1, \dots, t_n]$ is equivalent to the datum of n sections of $\mathcal{O}_{Y'}$), on the other hand one has

$$(1.3) \quad (Y[t_1, \dots, t_n])[t_{n+1}, \dots, t_m] = Y[t_1, \dots, t_m],$$

by virtue of the analogous formula for polynomial rings over $\mathbf{Z}$. The formula (1.2) implies that $Y[t_1, \dots, t_n]$ varies functorially with Y .

$Y[t_1, \dots, t_n]$ is of finite type and flat over Y .

DEFINITION 1.1. Let $f: X \rightarrow Y$ be a morphism, making X into a Y -prescheme. One says that f is *smooth*¹ at $x \in X$, or that X is *smooth over Y at x* , if there exists an integer $n \geq 0$, an open neighborhood U of x , and an étale Y -morphism from U to $Y[t_1, \dots, t_n]$. One says that f (resp. X) is *smooth* if it is smooth at every point of X . An algebra B over a ring A is said to be smooth at a prime ideal $\mathfrak{p}$ of B , if $\text{Spec}(B)$ is smooth over $\text{Spec}(A)$ at the point $\mathfrak{p}$; B is said to be smooth over A if it is smooth over A at every prime ideal $\mathfrak{p}$ of B . Finally, a local homomorphism $A \rightarrow B$ of local rings is said to be smooth (or B is said to be smooth over A)² if B is a localization of an algebra of finite type B_1 smooth over A .

One notes that the notion of smoothness of X over Y is local on X and on Y ; if X is smooth over Y , it is locally of finite type over Y .

PROPOSITION 1.1. *The set of points x of X at which f is smooth is open.*

This is trivial from the definition.

COROLLARY 1.2. *If B is smooth over A at $\mathfrak{p}$, then it is smooth over A at $\mathfrak{q}$ for every prime ideal $\mathfrak{q}$ of B contained in $\mathfrak{p}$.*

1.1 also implies that the last two definitions 1.1 coincide in their common domain of existence.

¹Old terminology: f is *simple* at x , or x is a *simple* point for f . This terminology led to confusion in various contexts (simple algebras, simple groups) and had to be abandoned.

²It is then better to say, as in EGA IV 18.6.1, that B is “*essentially smooth*” over A .

PROPOSITION 1.3. (i) *An étale morphism, in particular an open immersion, an identity morphism, is smooth.* (ii) *An extension of the base in a smooth morphism gives a smooth morphism.* (iii) *The composite of two smooth morphisms is smooth.*

(i) is trivial from the definition, more precisely one has:

COROLLARY 1.4. *étale = quasi-finite + smooth.*

(ii) follows at once from the analogous fact for étale morphisms (I 4.6) and for the projections $Y[t_1, \dots, t_n] \rightarrow Y$ (cf. (1.2)). For (iii), this follows formally from the fact that it is true separately for “étale” (I 4.6) and for projections of the type $Y[t_1, \dots, t_n] \rightarrow Y$ (cf. (1.3)), and from the two facts cited for (ii): Suppose Y is smooth over Z and X is smooth over Y , let us prove that X is smooth over Z ; one may suppose Y étale over $Z[t_1, \dots, t_n]$ and X étale over $Y[s_1, \dots, s_m]$, the first hypothesis therefore implies that $Y[s_1, \dots, s_m]$ is étale over $Z[t_1, \dots, t_n][s_1, \dots, s_m] = Z[t_1, \dots, s_m]$, hence X is étale over $Z[t_1, \dots, s_m]$, qed.

REMARK 1.5. The integer n appearing in def. 1.1 is well determined, for one sees at once that it is the dimension of the local ring of x in its fiber $f^{-1}(f(x))$. One calls it the “relative dimension” of X over Y . It behaves additively for the composition of morphisms.

2. Some criteria of smoothness of a morphism

THEOREM 2.1. *Let $f: X \rightarrow Y$ be a morphism locally of finite type, let $x \in X$ and $y = f(x)$. For f to be smooth at x , it is necessary and sufficient that (a) f be flat at x , and (b) $f^{-1}(y)$ be smooth over $k(y)$ at x .*

The composite of two flat morphisms being flat, and $Y[t_1, \dots, t_n] \rightarrow Y$ being a flat morphism, one sees that smooth implies flat; taking 1.3 (ii) into account this proves necessity. Suppose (a) and (b) are satisfied, let V be an affine neighborhood of y with ring A , U an affine neighborhood of x over V , with ring B . Taking U small enough, one may suppose by (b) that there exists an étale $k(y)$ -morphism

$$g: U|f^{-1}(y) \longrightarrow \operatorname{Spec} k[t_1, \dots, t_n] \quad (k = k(y))$$

defined by n sections g_i of the structure sheaf of $U|f^{-1}(y)$. One easily sees that one may suppose that the g_i (which a priori are elements of $B \otimes_A k = BS^{-1}$, where $S = A - \mathfrak{p}$, $\mathfrak{p}$ the prime ideal of A corresponding to y) come from sections of the structure sheaf of U , hence that g is induced by a morphism, still denoted g

$$g: U \longrightarrow Y[t_1, \dots, t_n]$$

(up to multiplying the g_i by one and the same nonzero element of k). Now U is flat over Y by (a), the same is true of $Y[t_1, \dots, t_n]$, on the other hand g induces an étale morphism between the fibers over y , hence g is étale at x by (I 5.8), qed.

COROLLARY 2.2. *Let S be a prescheme, $f: X \rightarrow Y$ an S -morphism of finite type, Y being of finite type and flat over S , $x \in X$, s the projection of x on S . For f to be smooth at x , it is necessary and sufficient that X be flat (or again: smooth) over S at x , and that the morphism $f_s: X_s \rightarrow Y_s$ induced on the fibers of s be smooth at x .*

Only sufficiency requires a proof, and follows from the criterion 2.1, together with the flatness criterion (I 5.9).

To state the following result, “recall” that a morphism $f: X \rightarrow Y$ locally of finite type is said to be *equidimensional* at the point $x \in X$ if (setting $y = f(x)$) one can find an open neighborhood U of x , every component of which dominates a component of Y such that, for every $y' \in Y$, the irreducible components of $f^{-1}(y') \cap U$ all have one and the same dimension independent of y' . It is enough, moreover, in this condition to take for y' the generic points of the irreducible components of Y passing through y , and the point y . If for example X and Y are integral and f is dominant, the condition means that the components of the $f^{-1}(y)$ passing through x have “the right” dimension, i.e. the dimension of the generic fiber (recall

that they are always $\geq$ the dimension of the generic fiber). If f is equidimensional at x , the dimension of its fiber at x being n , and if $g: U \rightarrow Y' = Y[t_1, \dots, t_n]$ is a Y -morphism of a neighborhood U of x , inducing a morphism on the fibers of y which is quasi-finite at x (or again, which amounts to the same, if g is quasi-finite at x), then one shows that every irreducible component of U passing through x dominates an irreducible component of Y' . Moreover by virtue of the “normalization lemma”, such a g always exists (and conversely, if there exists a quasi-finite Y -morphism g of an open neighborhood U of x into a Y -scheme of the form $Y' = Y[t_1, \dots, t_n]$, such that every component of U passing through x dominates a component of Y' , then f is equidimensional at x). This being so:

PROPOSITION 2.3. *Let $f: X \rightarrow Y$ be a morphism locally of finite type, x a point of X , $y = f(x)$, one assumes $\mathcal{O}_y$ normal. For f to be smooth at x , it is necessary and sufficient that f be equidimensional at x , and that $f^{-1}(y)$ be smooth over $k(y)$ at x .*

One sees at once from the definition that a smooth morphism is equidimensional (N.B. a flat morphism of finite type is not necessarily equidimensional at x , even if its fiber at x is irreducible). Let us prove the converse. Since $f^{-1}(y)$ is smooth over $k(y)$ at x , one may suppose (replacing X if need be by a suitable neighborhood of x) that there exists a Y -morphism

$$g: X \longrightarrow Y[t_1, \dots, t_n] = Y'$$

inducing an étale morphism on the fibers of y , and a fortiori quasi-finite at x . Hence g is unramified, and (f being equidimensional at x) the irreducible components of X passing through x each dominate a component of Y' , a fortiori the homomorphism $\mathcal{O}_{y'} \rightarrow \mathcal{O}_x$ deduced from g (where $y' = g(x)$) is *injective*. This homomorphism is moreover unramified, and $\mathcal{O}_{y'}$ is normal since it is a localization of the ring $\mathcal{O}_y[t_1, \dots, t_n]$, which is normal since $\mathcal{O}_y$ is so. Hence the homomorphism $\mathcal{O}_{y'} \rightarrow \mathcal{O}_x$ is étale (I 9.5 (ii)).

REMARKS 2.4. The preceding statement still holds upon replacing the hypothesis that $\mathcal{O}_y$ is normal by the weaker hypothesis that Y is *geometrically unibranch* at y , (cf. I 11) — since (I 9.5) holds under this hypothesis. Let us take this opportunity to note at the same time that if the residue field of a local integral domain A is algebraically closed, then analytically integral (i.e. $\hat{A}$ is an integral domain) implies geometrically unibranch, the converse being moreover true in any category of “good rings”, more precisely in a category of rings stable under the usual operations, and in which the completion of a normal local ring is normal (a condition fulfilled, by virtue of the “analytic normality theorem” of Zariski, in the category of affine algebras and their localizations)³.

“Recall” finally in the present context the following result, due to Hironaka⁴ which sometimes allows one to ensure that $f^{-1}(y)$ is a reduced scheme, i.e. that it is also what many algebraic geometers abusively regarded as the fiber (without multiplicity) of f above x (namely $f^{-1}(y)_{\text{red}}$):

PROPOSITION 2.5. *Let $f: X \rightarrow Y$ be a dominant morphism of finite type of reduced preschemes, y a point of Y such that $\mathcal{O}_y$ is regular. One assumes that all the components of $f^{-1}(y)$ are of multiplicity 1 (cf. the definition below), and that $f^{-1}(y)_{\text{red}}$ is normal. Then $f^{-1}(y)$ is reduced hence normal, X is normal at all the points of $f^{-1}(y)$, finally X is flat over Y at all the points of $f^{-1}(y)$.*

One says that a component Z of $f^{-1}(y)$ is of *multiplicity 1* if, x denoting the generic point of Z , one has (i) $\dim \mathcal{O}_x = \dim \mathcal{O}_y$ (i.e. Z is not an “excess component”, that is, is not “of too large dimension”; (ii) the maximal ideal of $\mathcal{O}_x$ is generated by the maximal ideal of $\mathcal{O}_y$ (which a priori, by virtue of the choice of x , generates an ideal of definition of $\mathcal{O}_x$).

Taking 2.3 or 2.1 into account one thus finds:

³Cf. EGA IV 7.8.

⁴Cf. EGA IV 5.12.10.

COROLLARY 2.6. *Let $f: X \rightarrow Y$ be a dominant morphism of finite type of reduced preschemes, y a point of Y such that $\mathcal{O}_y$ is regular. For f to be smooth at the points of X above y , it is necessary and sufficient that the components of $f^{-1}(y)$ be of multiplicities 1, and that $f^{-1}(y)_{\text{red}}$ be smooth over $k(y)$.*

This situation was considered especially in the past when Y was the spectrum of a discrete valuation ring A , and was commonly designated by phrases such as: “if the reduction of X with respect to the given valuation is nice”... Moreover, X then designated a closed subscheme (if one may say so) of a $\mathbf{P}_K^n$ (K being the field of fractions of A) and for want of an adequate language, the more intrinsic role of an object “defined over A ” (and not merely over K) hardly appeared.

3. Permanence properties

PROPOSITION 3.1. *Let $f: X \rightarrow Y$ be a morphism, let $x \in X$ and $y = f(x)$. Suppose f is smooth at x . For $\mathcal{O}_x$ to be reduced (resp. regular, resp. normal) it is necessary and sufficient that $\mathcal{O}_y$ be so.*

This statement is indeed known when X is of the form $Y[t_1, \dots, t_n]$, and it is proved in (I, no. 9) for an étale morphism; the general case is deduced from this at once thanks to definition 1.1.

We do not spell out here the other permanence properties, already resulting from flatness alone, or from the fact that X is locally quasi-finite and flat over a Y -prescheme of the form $Y[t_1, \dots, t_n]$ (or, as we shall say, that X is Cohen–Macaulay over Y). Let us only note that from this last fact it follows that

$$(3.1) \quad \dim \mathcal{O}_x = \dim \mathcal{O}_y + n - d, \quad \text{prof } \mathcal{O}_x = \text{prof } \mathcal{O}_y + n - d,$$

where n is the dimension of the fiber of f at x , and d the transcendence degree of $k(x)$ over $k(y)$, whence (setting $\text{coprof} = \dim - \text{prof}$)⁵

$$(3.2) \quad \text{coprof } \mathcal{O}_x = \text{coprof } \mathcal{O}_y.$$

It follows for example that $\mathcal{O}_x$ is Cohen–Macaulay (resp. without embedded components) if and only if the same is true of $\mathcal{O}_y$.

4. Differential properties of smooth morphisms

To simplify, we shall restrict ourselves for the most part to differential calculus of order 1, confining ourselves to brief indications for higher order (where the results are just as simple).

For the definition of the sheaf of 1-differentials $\Omega_{X/Y}^1$ of a Y -prescheme X , cf. (I no. 1). Suppose that X and Y are S -preschemes, the structure morphism $f: X \rightarrow Y$ being an S -morphism. Then f defines a homomorphism of Modules (compatible with f)

$$(4.1) \quad f^*: \Omega_{Y/S}^1 \longrightarrow \Omega_{X/S}^1$$

in other words, $\Omega_{X/S}^1$ is *contravariant* in the S -prescheme X . Moreover (4.1) is equivalent to a homomorphism of Modules on X

$$(4.1\text{bis}) \quad f^*(\Omega_{Y/S}^1) \longrightarrow \Omega_{X/S}^1$$

also denoted by f^* for want of a better notation, and which fits into a canonical exact sequence of homomorphisms of Modules

$$(4.2) \quad f^*(\Omega_{Y/S}^1) \longrightarrow \Omega_{X/S}^1 \longrightarrow \Omega_{X/Y}^1 \longrightarrow 0$$

All these homomorphisms are defined by the condition of being of local nature (which reduces one to the affine case) and of commuting with the operators d . The exactness of (4.2) is

⁵For these formulas, cf. EGA IV 6.1 and 6.3.

classical and trivial, and transcribes in the affine case as the exact sequence (corresponding to a homomorphism $B \rightarrow C$ of A -algebras):

$$(4.2\text{bis}) \quad \Omega_{B/A}^1 \otimes_B C \longrightarrow \Omega_{C/A}^1 \longrightarrow \Omega_{C/B}^1 \longrightarrow 0$$

LEMMA 4.1. *Let $f: X \rightarrow Y$ be a morphism of S -preschemes. If f is unramified (resp. étale) then $f^*(\Omega_{Y/S}^1) \rightarrow \Omega_{X/S}^1$ is surjective (resp. an isomorphism). The converse is true in the “unramified” case, if f is assumed locally of finite type.*

The unramified case follows from the exact sequence (4.2) and from (I 3.1), but can also be seen directly as the étale case. Consider the diagram

$$\begin{array}{ccccc} X & \xrightarrow{\Delta_{X/Y}} & X \times_Y X & \longrightarrow & X \times_S X \\ & & \downarrow & & \downarrow \\ & & Y & \xrightarrow{\Delta_{Y/S}} & Y \times_S Y \end{array}$$

in which $X \times_Y X$ is identified with the fibered product of Y and $X \times_S X$ over $Y \times_S Y$. Since f is unramified, $X \rightarrow X \times_Y X$ is an open immersion, hence the “conormal” sheaf of the composite immersion $\Delta_{X/S}$ of the latter with $X \times_Y X \rightarrow X \times_S X$ is isomorphic to the inverse image on X of the conormal sheaf for the immersion $X \times_Y X \rightarrow X \times_S X$. On the other hand, $X \rightarrow Y$ being étale hence flat, $X \times_S X \rightarrow Y \times_S Y$ is flat, hence the conormal sheaf for the immersion $X \times_Y X \rightarrow X \times_S X$ is isomorphic to the inverse image of the conormal sheaf for the immersion $Y \rightarrow Y \times_S Y$, i.e. the inverse image of $\Omega_{Y/S}^1$. The conclusion follows from this.

LEMMA 4.2. *Let $X = Y[t_1, \dots, t_n]$, Y being an S -prescheme. Then the sequence of canonical homomorphisms*

$$0 \longrightarrow f^*(\Omega_{Y/S}^1) \longrightarrow \Omega_{X/S}^1 \longrightarrow \Omega_{X/Y}^1 \longrightarrow 0$$

is exact and $\Omega_{X/Y}^1$ is free with basis the $d_{X/Y}t_i$.

The verification (purely affine) is immediate. (N.B. one already knows the exactness of (4.2)).

Combining these two statements and definition 1.1, one finds

THEOREM 4.3. *Let $f: X \rightarrow Y$ be a smooth morphism of S -preschemes, then:*

(i) *The sequence of canonical homomorphisms*

$$0 \longrightarrow f^*(\Omega_{Y/S}^1) \longrightarrow \Omega_{X/S}^1 \longrightarrow \Omega_{X/Y}^1 \longrightarrow 0$$

is exact.

(ii) *$\Omega_{X/Y}^1$ is locally free, its rank n at x is equal to the relative dimension of f at x .*

COROLLARY 4.4. *The homomorphism $f^*(\Omega_{Y/S}^1) \rightarrow \Omega_{X/S}^1$ is injective, its image in $\Omega_{X/S}^1$ is locally a direct factor.*

Let $u: F \rightarrow G$ be a homomorphism of Modules on the prescheme X ; one says that it is *universally injective* at $x \in X$ if the homomorphism $F_x \rightarrow G_x$ of $\mathcal{O}_x$ -modules is injective, and remains so upon tensoring with any $\mathcal{O}_x$ -algebra (or, which comes to the same thing moreover, with any $\mathcal{O}_x$ -module). It suffices for example that there exist an open neighborhood U of x such that u induces an isomorphism of $F|U$ onto a direct factor of $G|U$; this condition is also necessary when F and G are free (and of finite type) in a neighborhood of x ; more precisely in this case the following conditions are equivalent:

- (i) u is injective at x and $\text{Coker } u$ free at x ;
- (ii) There exists an open neighborhood U of x such that u induces an isomorphism of $F|U$ onto a direct factor of $G|U$;
- (iii) u is universally injective at x ;

- (iv) the homomorphism $F_x \otimes k(x) \rightarrow G_x \otimes k(x)$ on the restricted “fibers” induced by u is injective;
- (v) The transposed homomorphism $\check{G} \rightarrow \check{F}$ is surjective at the point x (or again, which comes to the same thing, in a neighborhood of x).

(Circular proof, (iv) $\Rightarrow$ (v) follows from Nakayama, on the other hand (v) $\Rightarrow$ (i) since a locally free quotient sheaf is necessarily a direct factor). Geometrically, the situation envisaged means that u corresponds to an isomorphism of the vector bundle whose sheaf of sections is F , onto a subbundle of the analogous vector bundle defined by G . Of course, it does not suffice for this that $F \rightarrow G$ be injective.

COROLLARY 4.5. *Let $f: X \rightarrow Y$ be a morphism of S -preschemes, locally of finite type, $x \in X$, $y = f(x)$, s the projection of x and y on S . One assumes Y is smooth at y over S . Equivalent conditions:*

- (i) f is smooth at x .
- (ii) X is smooth over S at x , and $f^*(\Omega_{Y/S}^1) \rightarrow \Omega_{X/S}^1$ is universally injective at x , i.e. it is an injective homomorphism at x and its cokernel $\Omega_{X/Y}^1$ is free at x .

Necessity follows from 1.3 (iii) and from 4.3 (i) (ii); let us prove sufficiency. Since the dg ($g \in \mathcal{O}_x$) generate the module $\Omega_{X/Y}^1$ at x , one can find g_i ($1 \leq i \leq n$) such that the images of the dg_i in $(\Omega_{X/Y}^1)_x$ form a basis of this module. Taking X small enough, one may assume that the g_i come from sections of $\mathcal{O}_X$, and thus define a Y -morphism $g: X \rightarrow Y' = Y[t_1, \dots, t_n]$. Using the hypothesis and lemma 4.2, one sees easily that the corresponding homomorphism $g^*(\Omega_{Y'/S}^1) \rightarrow \Omega_{X/S}^1$ is bijective at x , which reduces us to proving the

COROLLARY 4.6. *Let $f: X \rightarrow Y$ be a morphism of smooth S -preschemes. For f to be étale at $x \in X$, it is necessary and sufficient that $f^*(\Omega_{Y/S}^1) \rightarrow \Omega_{X/S}^1$ be an isomorphism at x .*

One knows that this is necessary by 4.1, and this condition implies that f is unramified at x by the same lemma. By virtue of 2.2, one is reduced to the case where $S = \text{Spec}(k)$. Since Y is smooth over k , it is regular, hence a fortiori normal, and by virtue of (I 9.5 (ii)) one is reduced to proving that $\mathcal{O}_y \rightarrow \mathcal{O}_x$ is injective, or equivalently that $\mathcal{O}_y$ and $\mathcal{O}_x$ have the same dimension. Now these dimensions are respectively the ranks of $\Omega_{Y/k}^1$ and $\Omega_{X/k}^1$ at y resp. x , hence equal by virtue of the hypothesis.

REMARKS 4.7. X and Y being assumed smooth over S , the criterion 4.5 (ii) of smoothness of $f: X \rightarrow Y$ can also be stated by saying that for every $x \in X$, the *tangent map* (relative to the base S) of f at x , i.e. the transpose of the homomorphism of the $k(x)$ finite-dimensional vector spaces, restricted fibers of $f^*(\Omega_{Y/S}^1)$ and $\Omega_{X/S}^1$ at x , is *surjective*. This is a quite familiar hypothesis in particular among people working with analytic spaces. The non-singularity hypothesis which they ordinarily make (which means that X and Y are “smooth over $\mathbf{C}$ ”, cf. no. 5) seems due only to the fear which singular points of algebraic varieties or analytic spaces still inspire in many geometers.

Let us point out the following particular case of 4.6:

COROLLARY 4.8. *Let X be an S -prescheme, $g: X \rightarrow S[t_1, \dots, t_n]$ an S -morphism, defined by the sections g_i ($1 \leq i \leq n$) of $\mathcal{O}_X$, x a point of X such that X is smooth over S at x . For g to be étale at x , it is necessary and sufficient that the dg_i ($1 \leq i \leq n$) form a basis of $\Omega_{X/S}^1$ at x (or, which comes to the same thing, that their images in $\Omega_{X/S}^1(x) = (\Omega_{X/S}^1)_x \otimes_{\mathcal{O}_x} k(x)$ form a basis of this vector space over $k(x)$).*

Let X be a prescheme, Y a closed subprescheme of X defined by a coherent sheaf $\mathcal{J}$ of ideals. Thus $\mathcal{J}/\mathcal{J}^2$ may be regarded as a coherent sheaf on Y (the *conormal sheaf* of Y in X). If now X is an S -prescheme, one has a canonical exact sequence of quasi-coherent sheaves on Y

$$(4.3) \quad \mathcal{J}/\mathcal{J}^2 \xrightarrow{d} \Omega_{X/S}^1 \otimes_{\mathcal{O}_X} \mathcal{O}_Y \longrightarrow \Omega_{Y/S}^1 \longrightarrow 0$$

whose right-hand part is none other than (4.2) (with the role of X and Y interchanged, taking into account that $\Omega_{Y/X}^1 = 0$), while the homomorphism $\mathcal{J}/\mathcal{J}^2 \rightarrow \Omega_{X/S}^1 \otimes_{\mathcal{O}_X} \mathcal{O}_Y$ is deduced from the homomorphism (in general not linear) $g \rightarrow dg$ by passing to quotients. The exactness of (4.3) is classical and moreover trivial, and is interpreted in the affine case by the following exact sequence (corresponding to a surjective homomorphism $B \rightarrow C$ of A -algebras, with kernel J):

$$(4.3\text{bis}) \quad J/J^2 \longrightarrow \Omega_{B/A}^1 \otimes_B C \longrightarrow \Omega_{C/A}^1 \longrightarrow 0 \quad (C = B/J)$$

(an exact sequence which had already been used implicitly in the proof of (I 7.2)!).

PROPOSITION 4.9. *Let X be an S -prescheme, Y a closed subscheme of X defined by a coherent sheaf $\mathcal{J}$ of ideals on X , x a point of X , g_i ($1 \leq i \leq n$) sections of $\mathcal{O}_X$, defining an S -morphism*

$$g: X \longrightarrow S[t_1, \dots, t_n] = X'$$

finally let p be an integer, $0 \leq p \leq n$. Suppose X is smooth over S at x . The following conditions are equivalent:

- (i) *There exists an open neighborhood X_1 of x in X such that $g|_{X_1}$ is étale and $Y_1 = Y \cap X_1$ (trace of Y on X_1) is the inverse image of the closed subscheme $Y' = S[t_{p+1}, \dots, t_n]$ of $X' = S[t_1, \dots, t_n]$ (i.e. the g_i ($1 \leq i \leq p$) generate $\mathcal{J}|_{X_1}$):*

$$\begin{array}{ccc} Y_1 & \xrightarrow{\quad} & X_1 \\ \downarrow & & \downarrow \text{étale} \\ Y' = S[t_{p+1}, \dots, t_n] & \longrightarrow & X' = S[t_1, \dots, t_n] \end{array}$$

- (ii) *Y is smooth over S at x , the g_i ($1 \leq i \leq p$) define elements of $\mathcal{J}_x$, the $dg_i(x)$ ($1 \leq i \leq n$) form a basis of $\Omega_{X/S}^1(x)$ over $k(x)$, the $dg'_i(x)$ ($p+1 \leq i \leq n$) form a basis of $\Omega_{Y/S}^1(x)$ over $k(x)$ (where the g'_i denote the restrictions of the g_i to Y ; the differentials are taken with respect to S).*
- (iii) *The g_i ($1 \leq i \leq p$) define a system of generators of $\mathcal{J}_x$, and the $dg_i(x)$ ($1 \leq i \leq n$) form a basis of $\Omega_{X/S}^1(x)$ over $k(x)$.*
- (iv) *Y is smooth over S at x , the g_i ($1 \leq i \leq p$) form a minimal system of generators of $\mathcal{J}_x$, the $dg'_i(x)$ ($p+1 \leq i \leq n$) form a basis of $\Omega_{Y/S}^1(x)$ over $k(x)$.*

Moreover, under these conditions, $\mathcal{J}/\mathcal{J}^2$ is a free Module on Y at x , admitting as basis at x the classes of the g_i ($1 \leq i \leq p$), and the canonical homomorphism $\mathcal{J}/\mathcal{J}^2 \rightarrow \Omega_{X/S}^1 \otimes \mathcal{O}_Y$ is universally injective at x .

REMARK. This implies that p is well determined by the other conditions, either as the rank of the free Module $\mathcal{J}/\mathcal{J}^2$ on Y at x , or again as the minimum number of generators of $\mathcal{J}_x$ on X , or finally by the fact that the relative dimension of Y rel. S at x is $n - p$.

Proof. Suppose first that (i) is satisfied. Then by (I 4.6 (iii)) Y_1 is étale over Y' , hence by definition it is smooth over S at x (of relative dimension $n - p$), hence the same is true of Y . It then follows from (4.8) that the dg_i ($1 \leq i \leq n$) form a basis of $\Omega_{X/S}^1$ at x , and that the dg'_i ($p+1 \leq i \leq n$) a basis of $\Omega_{Y/S}^1$ at x , whence it follows by the exact sequence (4.3) that the g_i ($1 \leq i \leq p$) are linearly independent in $\mathcal{J}/\mathcal{J}^2$ (regarded as a Module on Y) at x ; since the g_i ($1 \leq i \leq p$) generate $\mathcal{J}_x$, it follows that the $g_i \bmod \mathcal{J}_x^2$ form a basis of $\mathcal{J}/\mathcal{J}^2$ at x . This implies on the one hand that the g_i ($1 \leq i \leq p$) form a minimal system of generators of $\mathcal{J}_x$, on the other hand that the homomorphism $\mathcal{J}/\mathcal{J}^2 \rightarrow \Omega_{X/S}^1 \otimes \mathcal{O}_Y$ of (4.3) is universally injective at x (for it maps a basis of a free Module at x onto a part of a basis of a free Module at x — N.B. these are Y -Modules). This proves that (i) implies (ii), (iii), (iv), as well as the last assertions of proposition 4.9.

(iii) implies (i) by virtue of corollary 4.8.

(ii) implies (i). Indeed, the first hypothesis in (ii) means that (replacing if necessary X by an open neighborhood of x in X) g induces a morphism $h: Y \rightarrow Y'$. By 4.8, the other two hypotheses of (ii) mean that g is étale at x , and h étale at x . Let then Y'' be the inverse image of Y' by g . Thus Y is a closed subprescheme of Y'' , which is étale over Y' at x by (I 4.6 (iii)) since g is étale at x . Hence the immersion morphism $Y \rightarrow Y''$ is itself étale (I 4.8) hence an open immersion (I 5.8 or I 5.2), hence replacing still X by a suitable open neighborhood X_1 of x , one obtains (i).

What precedes establishes the equivalence of conditions (i), (ii), (iii), and the fact that they imply (iv), it remains to prove that (iv) $\Rightarrow$ (ii), which is immediate (taking into account that $\Omega_{X/S}^1$ is free on X at x) once one knows that the fact that Y is smooth over S at x implies that $\mathcal{J}/\mathcal{J}^2$ is free on Y at x , and the homomorphism $\mathcal{J}/\mathcal{J}^2 \rightarrow \Omega_{X/S}^1 \otimes \mathcal{O}_Y$ universally injective at x . This last point is included in the

THEOREM 4.10. *Let X be a smooth S -prescheme, Y a closed subprescheme of X defined by a coherent sheaf $\mathcal{J}$ of ideals on X , x a point of X . The following conditions are equivalent:*

- (i) Y is smooth over S at x .
- (ii) *There exists an open neighborhood X_1 of x in X and an étale S -morphism*

$$g: X_1 \longrightarrow X' = S[t_1, \dots, t_n]$$

such that $Y_1 = Y \cap X_1$ (trace of Y on X_1) is the subprescheme of X_1 inverse image by g of the closed subprescheme $Y' = S[t_{p+1}, \dots, t_n]$ of $X' = S[t_1, \dots, t_n]$, for a suitable p .

- (iii) *There exist generators g_i ($1 \leq i \leq p$) of $\mathcal{J}_x$, such that the dg_i form part of a basis of $\Omega_{X/S}^1$ at x (or, what amounts to the same, such that the $dg_i(x)$ in $\Omega_{X/S}^1$ are linearly independent over $k(x)$).*
- (iv) *The sheaf $\mathcal{J}/\mathcal{J}^2$ is free on Y at x , and the canonical homomorphism*

$$d: \mathcal{J}/\mathcal{J}^2 \longrightarrow \Omega_{X/S}^1 \otimes \mathcal{O}_Y$$

is universally injective at x ; or again: the sequence of canonical homomorphisms

$$0 \longrightarrow \mathcal{J}/\mathcal{J}^2 \longrightarrow \Omega_{X/S}^1 \otimes \mathcal{O}_Y \longrightarrow \Omega_{Y/S}^1 \longrightarrow 0$$

is exact at x , and $\Omega_{Y/S}^1$ is locally free at x .

Proof. One already knows that (ii) implies (i), (iii), (iv) by what precedes. Let us prove that (i) $\Rightarrow$ (ii) (which will at the same time complete the proof of 4.9). By theorem 4.3 (ii), the last two terms in the exact sequence (4.3) are free Modules on Y . Thus, since the images in $\Omega_{X/S}^1 \otimes_{\mathcal{O}_X} \mathcal{O}_Y$ of the dg ($g \in \mathcal{O}_X$) generate this module at x , hence their images in $\Omega_{Y/S}^1$ generate the latter at x , one can find g_i ($p+1 \leq i \leq n$) in $\mathcal{O}_X$ such that the dg'_i form a basis of $\Omega_{Y/S}^1$, then (by virtue of the exactness of (4.3)) complete the system of dg_i ($p+1 \leq i \leq n$) to a basis of the middle term by elements of the form dg_i ($1 \leq i \leq p$) where the g_i ($1 \leq i \leq p$) are in $\mathcal{J}_x$. The g_i come from sections of $\mathcal{O}_X$ on a neighborhood of x in X , which one may suppose equal to X . One is then under the conditions of 4.8 (ii), and one has established that this implies condition 4.8 (i), whence 4.10 (ii).

The implication (iii) $\Rightarrow$ (ii) in 4.10 follows at once from the implication (iii) $\Rightarrow$ (i) in 4.8. Hence (i), (ii), (iii) are equivalent, and imply (iv). Finally, it is trivial that (iv) implies (iii), taking into account that $g_i \in \mathcal{J}_x$ which form a basis of $\mathcal{J}_x \bmod \mathcal{J}_x^2$ generate $\mathcal{J}_x$ (Nakayama).

Moreover, the preceding proof shows the following:

COROLLARY 4.11. *Let X be an S -prescheme, Y a closed subprescheme defined by a coherent sheaf $\mathcal{J}$ of ideals on X , x a point of Y . Suppose X and Y are smooth over S at x . Let g_i be sections of $\mathcal{J}$ ($1 \leq i \leq p$). The following conditions are equivalent:*

- (i) *The g_i generate $\mathcal{J}_x$ and the $dg_i(x)$ are linearly independent in $\Omega_{X/S}^1(x)$ over $k(x)$.*
- (ii) *The $g_i \bmod \mathcal{J}^2$ form a basis of $\mathcal{J}/\mathcal{J}^2$ at x .*

- (iii) The g_i form a minimal system of generators of $\mathcal{J}_x$.
- (iv) One can find other sections g_i ($p+1 \leq i \leq n$) of $\mathcal{O}_X$ on a neighborhood X_1 of X , defining with the preceding ones an étale morphism $X_1 \rightarrow X' = S[t_1, \dots, t_n]$ such that $Y_1 = Y \cap X_1$ is the inverse image by g of the closed subprescheme $Y' = S[t_{p+1}, \dots, t_n]$ of $X' = S[t_1, \dots, t_n]$.

In particular:

COROLLARY 4.12. *Let X be an S -prescheme, F a section of $\mathcal{O}_X$, Y the subprescheme of zeros of F (defined by the coherent Ideal $F \cdot \mathcal{O}_X$), x a point of Y . Suppose X is simple over S at x . For Y to be smooth over S at x , it is necessary and sufficient that either F be zero in a neighborhood of x , or that $dF(x) \neq 0$ (where $dF(x)$ denotes the image of dF in the vector space $\Omega_{X/S}^1(x)$ over $k(x)$).*

This is sufficient by virtue of 4.10 criterion (iii). This is necessary, for since $\mathcal{J}$ is generated by one element, one must first have that $\mathcal{J}/\mathcal{J}^2$ at the point x be free of rank ≤ 1 . If this rank is 0, i.e. $\mathcal{J}/\mathcal{J}^2 = 0$ at x , it follows that $\mathcal{J} = 0$ at x by Nakayama, i.e. F is zero in a neighborhood of x . If this rank is 1, then F forms a minimal system of generators of $\mathcal{J}$ at x , and one concludes by (4.11), equivalence of (i) and (iii).

COROLLARY 4.13. *Let Y be an S -prescheme locally of finite type, S' a flat S -prescheme, $Y' = Y \times_S S'$, x' a point of Y' and x its canonical image in Y . For Y to be smooth over S at X , it is necessary and sufficient that Y' be smooth over S' at x' . In particular, if $S' \rightarrow S$ is flat and surjective, Y is smooth over S if and only if Y' is smooth over S' .*

Only sufficiency needs to be proved (necessity was seen in 1.3 (ii)). One may suppose (replacing Y by a suitable neighborhood of x , Y' by the inverse image of the latter) that Y is affine of finite type over S affine, hence Y is isomorphic to a closed subprescheme of a scheme $S[t_1, \dots, t_n]$. Consequently, Y' is identified with a closed subprescheme of $X' = X \times_S S'$. Since X is smooth over S , hence X' smooth over S' , one can apply the smoothness criteria 4.10. Here, criterion (iv) gives the result at once.

REMARKS 4.14. Criterion (iii) of Theorem 4.10 deserves to be called the *Jacobian criterion of smoothness*. It allows one to recognize, theoretically, whether a given S -prescheme Y is smooth over S at a point x of Y , since there always exists a neighborhood of Y isomorphic to a subprescheme of a smooth S -prescheme X , for example $X = S[t_1, \dots, t_n]$. It is moreover for $X = S[t_1, \dots, t_n]$, $S = \text{Spec}(A)$, that one usually states the Jacobian criterion (of course, in the classical case envisaged by Zariski, A was a field). One leaves to the reader to give the statement relative to the datum of an ideal J of $A[t_1, \dots, t_n]$ and of a prime ideal containing it, to which one is thus led. Let us note that it indeed seems at the present time (and especially since Nagata has succeeded in generalizing by non-differential methods Zariski's theorem stating that the set of regular points of an algebraic scheme is open) that the Jacobian criterion is scarcely of interest except in the form in which we give it here (i.e. using exclusively *relative* differentials and not *absolute* differentials, i.e. relative to the ring of absolute constants $\mathbf{Z}$). As so often, the consideration of differentials is more convenient here than that of derivations. Let us note finally that if Y is smooth over S at x , of relative dimension n , then there exists an open neighborhood of x in Y isomorphic to a subprescheme of $X = S[t_1, \dots, t_n]$ with $n = m + 1$, as follows from the definition and from (I 7.6).

Let A be a noetherian ring, x_i ($1 \leq i \leq n$) elements of A , J the ideal generated by the x_i . One says that the x_i form a *regular system of generators* of J if the canonical surjective homomorphism

$$(A/J)[t_1, \dots, t_n] \longrightarrow \text{gr}^J(A)$$

defined by the x_i (where the second member denotes the graded ring associated to A filtered by the powers of J) is an *isomorphism*. This condition also means that

- (i) The canonical surjective homomorphism

$$S_{A/J}(J/J^2) \longrightarrow \mathrm{gr}^J(A)$$

(where the first member denotes the symmetric algebra of the A/J -module J/J^2) is an isomorphism, and

- (ii)
- J/J^2
- is free and admits as basis the classes of the
- $x_i \bmod J^2$
- .

In this form, one sees that if $J \neq A$, the x_i form a *minimal system of generators* of J , and that *every other minimal system of generators of J is a regular system of generators* (N.B. “minimal” is taken in the strict sense: minimum number of elements, which is equivalent to the sense: minimal for inclusion, only if A is local); on the other hand, if $J = A$, every system of generators of J is regular.

The condition of regularity of a system of generators of an ideal is stable under localization with respect to an arbitrary multiplicatively closed set, and on the other hand one sees at once that for (x_i) to be a minimal system of generators of J , it already suffices that for every *maximal ideal* $\mathfrak{m}$ containing J , the x_i define a regular system of generators of $JA_{\mathfrak{m}}$ in $A_{\mathfrak{m}}$. This therefore reduces us to the case where A is a local ring with maximal ideal $\mathfrak{m}$, and where the x_i are in $\mathfrak{m}$. *Then the x_i form a regular system of generators of J if and only if they form an A -sequence in the sense of Serre⁶*, i.e. if for every i such that $1 \leq i \leq n$, x_i is not a divisor of 0 in $A/(x_1, \dots, x_{i-1})A$.

Finally, in the case where A is an algebra over a ring B , and where A/J is isomorphic as a B -algebra to B (so that J is the kernel of a homomorphism of B -algebras $A \rightarrow B$), then the x_i form a regular system of generators of J if and only if the canonical homomorphism

$$B[[t_1, \dots, t_n]] \longrightarrow \hat{A}$$

defined by the x_i (where the second member denotes the separated completion $\varprojlim A/J^{n+1}$ of A for the topology defined by the powers of J) is an *isomorphism* (it is in any case *surjective*).

All these facts are well known, and doubtless appear in Serre’s course on commutative algebra written up by Gabriel, up to minor details. (Where one finds N other characterizations of A -sequences, in the case where A is a local ring).

Let J be an ideal in a noetherian ring A . We shall say that J is a *regular ideal* if for every prime ideal $\mathfrak{p}$ of A , $JA_{\mathfrak{p}}$ admits a regular system of generators. It is obviously sufficient to verify this for $\mathfrak{p} \supset J$, and one may moreover restrict oneself to maximal $\mathfrak{p}$. More generally, let $\mathcal{J}$ be an Ideal on a locally noetherian prescheme X , one says that $\mathcal{J}$ is a *regular Ideal* if for every $x \in X$, $\mathcal{J}_x$ is an ideal of $\mathcal{O}_x$ which admits a regular system of generators. This is equivalent to the conjunction of the two following conditions:

- (a) The canonical surjective homomorphism

$$S_{\mathcal{O}_X/\mathcal{J}}(\mathcal{J}/\mathcal{J}^2) \longrightarrow \mathrm{gr}^{\mathcal{J}}(\mathcal{O}_X)$$

is an isomorphism and

- (b) The sheaf of
- $\mathcal{O}_X/\mathcal{J}$
- Modules
- $\mathcal{J}/\mathcal{J}^2$
- is locally free.

One then also says that the subscheme Y of X defined by $\mathcal{J}$ (hence such that $\mathcal{O}_Y$ extended by 0 is isomorphic to $\mathcal{O}_X/\mathcal{J}$) is *regularly immersed* in X , and one defines in the same way (in an evident manner) the notion of morphism of *regular immersion*, (resp. *regular at a point x*), immersion morphism $Y \rightarrow X$ identifying Y (resp. a suitable neighborhood of x) with a closed subscheme regularly immersed in an open subset of X . (One must not say: regular subscheme, for that would mean that the local rings of Y are regular). Finally, sections x_i of $\mathcal{J}$ are called a *regular system of generators* if for every $x \in X$, the corresponding elements of $\mathcal{O}_x$ form a regular system of generators of $\mathcal{J}_x$ (terminology compatible with that

⁶We shall now rather say “ A -regular sequence”, cf. EGA 0_{IV} 15.1.7 and 15.1.11.

introduced for generators of an ideal of a ring). This also means that the canonical surjective homomorphism

$$\mathcal{O}_Y[t_1, \dots, t_n] \longrightarrow \mathrm{gr}^{\mathcal{J}}(\mathcal{O}_X)$$

defined by the x_i is an isomorphism. If one knows in advance that the Ideal $\mathcal{J}$ is regular, this also means, simply, that at every point x of Y , the x_i define a *basis* of $\mathcal{J}/\mathcal{J}^2$ over $\mathcal{O}_{Y,x}$. (N.B. this condition is empty if Y is empty). Thus, for $\mathcal{J}$ to admit a regular system of generators, it is necessary and sufficient that $\mathcal{J}$ be regular, and that the $\mathcal{O}_Y$ -Module $\mathcal{J}/\mathcal{J}^2$ be globally free (and not only locally free), i.e. that the canonical homomorphism $S_{\mathcal{O}_Y}(\mathcal{J}/\mathcal{J}^2) \rightarrow \mathrm{gr}^{\mathcal{J}}(\mathcal{O}_X)$ be surjective, and that the $\mathcal{O}_Y$ -Module $\mathcal{J}/\mathcal{J}^2$ be globally free.

An *augmented ring* is said to be *regular* if the ideal of the augmentation is regular. Thus, if A is a local ring, regarded as augmented in its residue field k , then A is a regular local ring if and only if it is a regular augmented ring.

(To tell the truth, it seems that it was useless to begin by doing the preliminary sorites for rings; there is an advantage in beginning with sheaves at once. If one wants something in the noetherian case, it is the definition adopted here—a priori less strict than that by Serre's A -sequences—which seems preferable for the needs of differential calculus. Of course, to do things properly, one should also develop at least part of the theory of smooth morphisms in the non-noetherian setting⁷, probably starting from the Jacobian criterion, so as to obtain if possible all the essential formal properties of smooth morphisms and of étale morphisms i.e. smooth and quasi-finite; the converses alone appealing to noetherian hypotheses).

After these long terminological preliminaries, a little theorem:

THEOREM 4.15. *Let X be an S -prescheme locally of finite type, Y a closed subprescheme of X defined by a coherent sheaf $\mathcal{J}$ of ideals on X , x a point of X . One now assumes Y is smooth over S at x (and nothing on X). Then the following conditions are equivalent:*

- (i) X is smooth over S at x
- (ii) The immersion $i: Y \rightarrow X$ is regular at x , i.e. $\mathcal{J}_x$ is a regular ideal of $\mathcal{O}_x$.

COROLLARY 4.16. *Suppose Y is smooth over S . For X to be smooth over S in a neighborhood of Y (i.e. at the points of Y) it is necessary and sufficient that Y be regularly immersed in X , i.e. that the immersion $i: Y \rightarrow X$ be regular.*

Proof. (i) implies (ii). One applies 4.10 criterion (ii), since $g: X_1 \rightarrow X$ is *flat*, to show that the inverse image under g of the subprescheme Y' of X' is regularly immersed, one is reduced to proving that $Y' = S[t_{p+1}, \dots, t_n]$ is regularly immersed in $S[t_1, \dots, t_n]$, which is trivial (the t_i ($1 \leq i \leq p$) form a regular system of generators of the Ideal defining Y' in X').

(ii) implies (i). Let g_i ($1 \leq i \leq p$) be a regular system of generators of $\mathcal{J}_x$ and let g_i ($p+1 \leq i \leq n$) be elements of $\mathcal{O}_{X,x}$ such that their images g'_i in $\mathcal{O}_{Y,x}$ define an étale morphism

$$Y_1 \longrightarrow Y' = S[t_{p+1}, \dots, t_n]$$

of a neighborhoods Y_1 of Y in Y' . The g_i ($1 \leq i \leq n$) come from sections (of the same name) of $\mathcal{O}_X$ on a neighborhood X_1 of x , and one may suppose $X_1 = X$, $Y_1 = Y$. One thus obtains a morphism

$$g: X \longrightarrow X' = S[t_1, \dots, t_n]$$

and everything reduces to showing that this morphism is *étale* at x . Taking X_1 small enough, one may suppose that the g_i ($1 \leq i \leq p$) form a regular system of generators of $\mathcal{J}$ on all of X . In particular, they generate $\mathcal{J}$, hence the subprescheme Y of X is identified with the inverse image under g of the subprescheme Y' of X' . Let $x' = g(x)$, then the fiber of $X' \rightarrow X$ at x' is identical to the fiber of $Y \rightarrow Y'$ at x , hence is étale over $k(x')$, hence g is *unramified* at x , it remains to prove that g is *flat* at x . Now the associated graded of $\mathcal{O}_{X',x'}$ filtered by the powers of $\mathcal{J}'_x$ is *free* over $\mathcal{O}_{Y',x'}$ in all degrees, on the other hand the associated graded of $\mathcal{O}_{X,x}$ filtered by the powers of $\mathcal{J}_x = \mathcal{J}'_x \mathcal{O}_{X,x}$ is isomorphic (under the canonical homomorphism) to

⁷As is said in the foreword, this has now been done, cf. EGA IV 17, 18.

the tensor product of the former with $\mathcal{O}_{Y,x}$ (since both rings are polynomial rings in $n - p$ indeterminates, with ring of constants $\mathcal{O}_{Y',x'}$, resp. $\mathcal{O}_{Y,x}$), finally over $\mathcal{O}_{X',x'}/\mathcal{J}'_{x'} = \mathcal{O}_{Y',x'}$ $\mathcal{O}_{X,x}/\mathcal{J}_x = \mathcal{O}_{Y,x}$ is flat.

By a general flatness criterion (valid for a local homomorphism of noetherian local rings $A' \rightarrow A$, A' being equipped with an ideal $J' \neq A'$ such that the associated graded is free over A'/J' in every dimension) it follows that X is flat over X' at x , qed.

COROLLARY 4.17. *Let X be a prescheme locally of finite type over Y , i a section of X over Y , y a point of Y , $x = i(y)$, $\mathcal{J}$ the sheaf of ideals on X defined by the subprescheme $i(Y)$ (which we assume closed to simplify the statement, a condition satisfied if X is a scheme).*

The following conditions are equivalent:

- (i) X is smooth over Y at x
- (ii) i is a regular immersion at y
- (iii) The completed $\mathcal{O}_y$ -algebra of $\mathcal{O}_x$ for the topology defined by the powers of $\mathcal{J}_x$ is isomorphic to a formal power series algebra $\mathcal{O}_y[[t_1, \dots, t_n]]$.
- (iii bis) There exists an open neighborhood U of y such that the sheaf of algebras $\varprojlim i^*(\mathcal{O}_X/\mathcal{J}^{n+1})$ over $\mathcal{O}_Y$ is isomorphic to a sheaf of the form $\mathcal{O}_Y[[t_1, \dots, t_n]]$ over U .
- (iv) There exists an open neighborhood U of y , and an open neighborhood V of x , and finally a Y -morphism $g: V \rightarrow U[t_1, \dots, t_n]$, such that g is étale, that i induces a section of V over U , transformed by g into the zero section of $U[t_1, \dots, t_n]$ over U .

The equivalence of (i) and (ii) is a special case of theorem 4.15, setting $Y = S$, the equivalence of (ii) and (iii) (and morally of (ii) and (iii bis)) has been pointed out with the “recalls”. as for the equivalence of (i) and (iv), it is easily deduced from theorem 4.10 (equivalence of conditions (i) and (ii) of the latter).

COROLLARY 4.18. *Let X be a prescheme smooth over S . Then the diagonal morphism*

$$\Delta_{X/S}: X \longrightarrow X \times_S X$$

is a regular immersion, or as one also says, X is “differentially smooth” over S .

Indeed, this is a special case of corollary 4.16, since X and $X \times_S X$ are both smooth over S .

REMARKS 4.18. Let us recall (I 1) that if X is a prescheme over S , one introduces the quasi-coherent sheaves of algebras $\mathcal{P}_{X/S}^n = \mathcal{O}_{X \times_S X} / \mathcal{I}_X^{n+1}$ on X , (where $\mathcal{I}_X$ denotes the sheaf of ideals which defines the diagonal in $X \times_S X$), considered as a sheaf of $\mathcal{O}_X$ -algebras thanks to the first projection $\text{pr}_1: X \times_S X \rightarrow X$. The $\mathcal{P}_{X/S}^n$ form a projective system of Algebras over X , whose projective limit is denoted $\mathcal{P}_{X/S}^\infty$ and is none other than the structure sheaf of the formal completion of $X \times_S X$ along the diagonal (now assuming X is locally of finite type over S , hence the $\mathcal{P}_{X/S}^n$ are coherent). To say that X is differentially smooth over S (i.e. that the diagonal morphism $\Delta_{X/S}$ is a regular immersion) also means that $\mathcal{P}_{X/S}^\infty$ is regular, as a sheaf of algebras augmented to $\mathcal{O}_X$, i.e. that $\Omega_{X/S}^1$ is locally free and the canonical surjective homomorphism

$$S_{\mathcal{O}_X}(\Omega_{X/S}^1) \longrightarrow \text{gr}_*(\mathcal{P}_{X/S}^\infty)$$

is an isomorphism, or finally that every point of X has an open neighborhood on which the sheaf of augmented algebras $\mathcal{P}_{X/S}^\infty$ is isomorphic to a sheaf $\mathcal{O}_X[[t_1, \dots, t_n]]$.

Let s be a section of X over S , $\mathcal{J}$ the sheaf of ideals on X which it defines (assuming for simplicity that $s(S)$ is closed), one then has canonical isomorphisms of augmented $\mathcal{O}_X$ -algebras:

$$(4.4) \quad s^*(\mathcal{P}_{X/S}^n) = \mathcal{O}_X/\mathcal{J}^{n+1}, \quad s^*(\mathcal{P}_{X/S}^\infty) = \varprojlim_n \mathcal{O}_X/\mathcal{J}^{n+1}$$

These isomorphisms are functorial in an evident sense under change of base, and (taking this fact into account) give back a characterization of the sheaves of algebras $\mathcal{P}_{X/S}^n$ over S . If for example $S = \text{Spec}(k)$, k a field, then the datum of a section s of X over S is equivalent to the datum of a point x of X rational over k , and the preceding formulas mean that one has an isomorphism of k -algebras

$$(4.5) \quad \mathcal{P}_{X/S}^n(x) = \mathcal{O}_x / \mathfrak{m}_x^{n+1}$$

which justifies the name: “*sheaf of principal parts of order n on X rel. to S* ” given to $\mathcal{P}_{X/S}^n$. One sees moreover from (4.4) that *if X is differentially smooth over S at every point of $s(S)$, then X is smooth over S at every point of $s(S)$* , (corollary 4.17) *the converse being equally true* (corollary 4.18). Taking 4.13 into account, one easily concludes that if X is an S -prescheme locally of finite type, *X is smooth over S if and only if it is flat over S and differentially smooth over S* . (N.B. the flatness hypothesis is essential, as one sees by taking for X a closed subprescheme of S).

Let us note further, by way of recall, that one obtains a *second algebra structure* on $\mathcal{P}_{X/S}^n$ thanks to the projection $\text{pr}_2: X \times_S X \rightarrow X$, which is moreover deduced from the former by means of the *canonical involution* of the sheaf of rings $\mathcal{P}_{X/S}^n$, induced by the symmetry automorphism of $X \times_S X$. One denotes by $d_{X/S}^n$ or simply d^n , the homomorphism of sheaves of rings

$$(4.6) \quad d_{X/S}^n: \mathcal{O}_X \longrightarrow \mathcal{P}_{X/S}^n$$

which corresponds to this second Algebra structure. Taking the isomorphism (4.4) into account, this homomorphism transforms a section f of $\mathcal{O}_X$ into a section $d^n(f)$ of $\mathcal{P}_{X/S}^n$ whose inverse image under a section s of X over S is identified with the canonical image of f in $\Gamma(X, \mathcal{O}_X / \mathcal{I}^{n+1})$. This justifies the name “*system of principal parts of order n of f* ” given to $d^n f$, notably in the case where $S = \text{Spec}(k)$, envisaged in formula (4.5).

To finish, let us note that the homomorphism (4.6) may be considered as the *differential operator of order $\leq n$* ⁸ (relative to the prescheme of constants S) *universal* on $\mathcal{O}_X$, by agreeing to call differential operator of order $\leq n$ from $\mathcal{O}_X$ into a Module F , a homomorphism of sheaves D which factors as

$$D: \mathcal{O}_X \xrightarrow{d^n} \mathcal{P}_{X/S}^n \xrightarrow{u} F$$

where u is a homomorphism of $\mathcal{O}_X$ -Modules, moreover uniquely determined by D . This definition agrees with the intuitive recursive definition (D is a differential operator of order $\leq n$ if for every section g of $\mathcal{O}_X$ on an open set U of X , $f \mapsto D(fg) - D(f)g$ is a differential operator of order $\leq n-1$ on U). It follows that *if X is differentially smooth over S , the sheaf of rings of differential operators of all orders has the simple well-known structure* in differential calculus on differentiable manifolds, and in particular admits locally a basis of $\mathcal{O}_X$ -Module formed of the *divided powers* in commuting operators $\delta/\delta x_i$ ($1 \leq i \leq n$). If S is a sheaf of $\mathbf{Q}$ -algebras ($\mathbf{Q}$ = field of rationals) it suffices to take the ordinary polynomials in the $\delta/\delta x_i$. In this case moreover, and quite exceptionally, for X to be differentially smooth over S , it already suffices that $\Omega_{X/S}^1$ be locally free.

REMARK 4.19. The terminology “regular immersion”, “regular ideal”, etc. introduced in this no. has met with a rather lively and general opposition (Chevalley, Serre). One has proposed “Cohen–Macaulay ideal” or “Macaulay ideal” or “Macaulayan ideal”, which would morally also oblige one to adopt “Cohen–Macaulay immersion” or “Macaulay immersion”. This terminology however conflicts with another already used in future drafts of the multiplodogue, where a morphism (of finite type) is said to be “Cohen–Macaulay” at a point if it is flat at that point, and if the fiber passing through that point has there a local ring which is

⁸For everything concerning the present paragraph, one may consult EGA IV 16.8 to 16.12.

a Cohen–Macaulay ring. Until a satisfactory solution is found, we shall keep under all reservations the terminology introduced in this no.⁹

5. The case of a ground field

PROPOSITION 5.1. *Let k be a field, X a prescheme of finite type over k , x a point of X and n the dimension of X at x , $f: X \rightarrow \operatorname{Spec} k[t_1, \dots, t_n] = Y$ a morphism, defined by elements $f_i \in \Gamma(X, \mathcal{O}_X)$. The following conditions are equivalent (and imply that X is smooth over k at x , and a fortiori regular at x by 3.1):*

- (i) f is étale at x .
- (ii) The df_i form a basis of $\Omega_{X/k}^1$ at x .
- (iii) The df_i generate $\Omega_{X/k}^1$ at x .

Since (i) implies that X is smooth over k at x , the implication (i) $\Rightarrow$ (ii) is a special case of 4.8, (ii) $\Rightarrow$ (iii) is trivial, it remains to prove (iii) $\Rightarrow$ (i). Now if (iii) is satisfied, f is net at x by virtue of lemma 4.1, hence (replacing x by an open neighborhood) quasi-finite, hence dominant for reasons of dimensions. Since Y is regular, it follows that f is étale by (I 9.5 (ii)) or (I 9.11).

COROLLARY 5.2. *Under the preliminary conditions of 5.1, suppose that $k(x)$ is a finite separable extension of k , and that the f_i ($1 \leq i \leq n$) define elements of $\mathfrak{m}_x$. Then the preceding conditions are equivalent to*

- (iv) *The f_i form a system of generators of $\mathfrak{m}_x$ (or equivalently: the $f_i \bmod \mathfrak{m}_x^2$ form a basis of $\mathfrak{m}_x/\mathfrak{m}_x^2$ over $k(x)$).*

Indeed, (iv) $\Rightarrow$ (iii) by virtue of the exact sequence

$$(5.1) \quad \mathfrak{m}_x/\mathfrak{m}_x^2 \longrightarrow \Omega_{\mathcal{O}_x/k}^1 \longrightarrow \Omega_{k(x)/k}^1 \longrightarrow 0$$

and taking into account that $\Omega_{k(x)/k}^1 = 0$ since $k(x)$ is étale over k . On the other hand (ii) implies (iv), for since X and $\operatorname{Spec}(k(x))$ are smooth over k at x , one can in the preceding exact sequence put a 0 on the left by virtue of 4.10 (iv).

COROLLARY 5.3. *Let x be a point of X (of finite type over k). If X is smooth over k at x , then $\mathcal{O}_x$ is regular, the converse being true if $k(x)$ is a finite separable extension of k .*

Indeed, the converse follows from 5.2, by taking a regular system (f_i) of generators of $\mathfrak{m}_x$. (N.B. instead of 5.2, one can also invoke theorem 4.15). One concludes:

PROPOSITION 5.4. *Let X be a prescheme of finite type over k . If X is smooth over k , it is regular, the converse being true if k is perfect.*

For the converse, one notes that by virtue of 5.3, X is smooth over k at every closed point, hence everywhere since the set of points where it is smooth is open.

THEOREM 5.5. *Let X be a prescheme of finite type over k , x a point of X , n the dimension of X at x , k' a perfect extension of k . The following conditions are equivalent:*

- (i) X is smooth over k at x .
- (ii) $\Omega_{X/k}^1$ is free of rank n at x .
- (ii bis) $\Omega_{X/k}^1$ is generated by n elements at x .
- (iii) X is differentially smooth over k at x .
- (iv) There exists an open neighborhood U of x such that $U \otimes_k k'$ is regular (i.e. the local rings of its points are regular).

⁹It is the one adopted in EGA 0_{IV} 15.1.7.

One has (i) $\Rightarrow$ (ii) by 4.3 (ii), (ii) $\Rightarrow$ (ii bis) trivially and (ii bis) $\Rightarrow$ (i) thanks to 5.1. Since X is flat over k , one has (i) $\Leftrightarrow$ (iii) by virtue of 4.18. One has (i) $\Rightarrow$ (iv) since smooth is invariant under extension of the base and implies regular, and (iv) $\Rightarrow$ (i) for by Proposition 5.4, one sees that $U \otimes_k k'$ is simple over k' , hence U is simple over k by 4.13.

Taking for x the generic point of X assumed irreducible, one finds:

COROLLARY 5.6. *Let K be a local Artin ring which is a localization of an algebra of finite type over the field k (for example, K may be an extension of finite type of k), let n be the transcendence degree over K of its residue field. Equivalent conditions:*

- (i) K is a finite separable extension of a purely transcendental extension $k(t_1, \dots, t_n)$ of k .
- (ii) $\Omega_{X/k}^1$ is a K -module free of rank n .
- (ii bis) $\Omega_{X/k}^1$ is a K -module admitting n generators.
- (iii) The completion O' of $K \otimes_k K$ for the topology defined by the powers of the augmentation ideal $K \otimes_k K \rightarrow K$ is a “regular” augmented K -algebra, i.e. isomorphic to an algebra of formal series in K (N.B. If K is a field, this is equivalent to saying that O' is a regular local ring).
- (iv) K is a separable extension of k .

Indeed, one can always consider K as the local ring of the generic point of an irreducible scheme of finite type X over k , and the conditions envisaged are the conditions of the same name in 5.5, taking in (iv) for k' an algebraically closed extension of k . Only the implication K separable over $k \Rightarrow X$ smooth over k at x , requires a proof. Now one is immediately reduced thanks to 4.13 to the case where the ground field is k' , hence algebraically closed, hence where there exists a point a of X rational over k . But then X is smooth over k at a by 5.4, a fortiori it is smooth over k at the generic point x , qed¹⁰.

One will note that in the case where K is an extension of finite type of k , the equivalence of (i), (ii), (ii bis), (iv) is well known, but that we have not made use of any of these already known equivalences. Of course, Proposition 5.1 contains as a special case that a sequence of elements x_i ($1 \leq i \leq n$) is a “separating transcendence basis” of K over k if and only if the dx_i form a basis of the K -module $\Omega_{K/k}^1$, a fact well known moreover.

COROLLARY 5.7. *Let X be a prescheme of finite type over a field k . For X to be smooth over k , it is necessary and sufficient that $\Omega_{X/k}^1$ be locally free, and that the local rings of the generic points of the irreducible components of X be separable extensions of k (this last condition being automatically satisfied if k is perfect and X reduced).*

One may assume X connected, let n be the rank of $\Omega_{X/k}^1$ assumed locally free. By the hypothesis and 5.6, this is also the transcendence degree of the extensions of k defined by the local rings of the generic points of X , hence all the irreducible components of X are of dimension n . One then concludes thanks to 5.5.

One should take care that if K is a finite extension (not necessarily separable) of k , then $\Omega_{K/k}^1$ is a free k -module, hence setting $X = \text{Spec}(K)$, $\Omega_{X/k}^1$ is a locally free sheaf, and X is reduced, without X being necessarily smooth over k . Extending then the scalars to the algebraic closure of k , one finds an analogous example, where k is algebraically closed, but X on the other hand not being reduced.

COROLLARY 5.8. *Let X be a prescheme of finite type over the field k , x a point of X , n the dimension of X at x , p the dimension of $\mathcal{O}_x$, i.e. the codimension in X of the closure Y of x in X ; hence $n - p$ the transcendence degree of $k(x)$ over k . Let f_i ($1 \leq i \leq n$) be elements of $\mathcal{O}_x$, such that $f_i \in \mathfrak{m}_x$ for $1 \leq i \leq p$. The following conditions are equivalent*

- (i) the germ of morphism at x

$$X \longrightarrow \text{Spec}(k[t_1, \dots, t_n])$$

¹⁰Cf. Errata at the end of the present Exp. II (p. 16)

- defined by the f_i is étale at x .
- (ii) The f_i ($1 \leq i \leq p$) generate $\mathfrak{m}_x$, i.e. form a regular system of parameters of $\mathcal{O}_x$, and the classes in $k(x)$ of the f_j ($p+1 \leq j \leq n$) form a separating transcendence basis, (i.e. the $d\bar{f}_j$ ($p+1 \leq j \leq n$) form a basis of $\Omega_{k(x)/k}^1$, or equivalently generate $\Omega_{k(x)/k}^1$).

Suppose (i) is satisfied. It follows that the $df_i(x)$ form a basis of $\Omega_{X/k}^1(x)$ (4.8) hence their images $d\bar{f}_i(x)$ in $\Omega_{k(x)/k}^1$ generate this vector space over k . Since the $\bar{f}_i$ for $1 \leq i \leq p$ are zero, it follows that it suffices to take the $d\bar{f}_i(x)$ with $p+1 \leq i \leq n$. Since the transcendence degree of $k(x)$ over k is $n-p$, it then follows from corollary 5.6 criterion (iii) (applied to $K = k(x)$) that Y is smooth over k at its generic point x , and that the $d\bar{f}_i(x)$ ($p+1 \leq i \leq n$) form a basis of $\Omega_{k(x)/k}^1$ over $k(x)$. Consequently, condition (ii) of 4.9 is satisfied, hence also condition (iii) and in particular the f_i ($1 \leq i \leq p$) form a system of generators of $\mathfrak{m}_x$. Since $\mathcal{O}_x$ is of dimension p , they therefore form a regular system of parameters at x . This proves (ii).

Suppose (ii) is satisfied. By virtue of the exact sequence (5.1), it follows that the $df_i(x)$ generate $\Omega_{X/k}^1$, whence (i) thanks to prop. 5.1.

COROLLARY 5.9. *Let X be a prescheme of finite type over the field k , x a point of X , n the dimension of X at x , p the dimension of $\mathcal{O}_x$, i.e. the codimension of the closure Y of x in X , hence $n-p$ the transcendence degree of $k(x)$ over k . Equivalent conditions:*

- (i) $\mathcal{O}_x$ is regular and $k(x)$ is a separable extension of k .
- (ii) X is smooth over k at x , and the canonical homomorphism

$$\mathfrak{m}_x/\mathfrak{m}_x^2 \longrightarrow \Omega_{\mathcal{O}_x/k}^1 \otimes_{\mathcal{O}_x} k(x) = \Omega_{X/k}^1(x)$$

is injective.

- (iii) There are $f_i \in \mathcal{O}_x$ ($1 \leq i \leq n$) with $f_i \in \mathfrak{m}_x$ for $1 \leq i \leq p$, such that the germ of morphism at x from X to $\text{Spec}(k[t_1, \dots, t_n])$ defined by the f_i is étale at x (i.e. by 5.1 such that the $df_i(x)$ generate $\Omega_{X/k}^1(x)$).
- (iv) There are $f_i \in \mathcal{O}_x$ ($1 \leq i \leq n$) such that the f_i ($1 \leq i \leq p$) generate $\mathfrak{m}_x$ and that the $df_j(x)$ ($p+1 \leq j \leq n$) generate $\Omega_{k(x)/k}^1$ over $k(x)$.

The equivalence of (iii) and (iv) follows from corollary 5.8, these conditions are also equivalent by 4.9 to the fact that X is smooth over k at x and that condition (ii) of 4.10 is satisfied. Hence they are equivalent to the fact that X is smooth over k at x and that condition (iv) of 4.10 is satisfied, hence to 5.9 (ii). Or to the fact that X is smooth over k at x and that condition (i) of 4.10 is satisfied, which here simply means that $k(x)$ is separable over k . This implies 5.9 (i), it remains to prove that 5.9 (i) implies it, i.e. to prove the

COROLLARY 5.10. *Let x be a point of a prescheme of finite type over the field k , such that $k(x)$ is separable over k . For X to be smooth over k at x , it is necessary and sufficient that it be regular at x (i.e. that the local ring $\mathcal{O}_x$ be regular).*

Indeed, if this is so, one can evidently find $f_i \in \mathcal{O}_x$ ($1 \leq i \leq n$) satisfying condition 5.9 (iv).

Errata. In the present no., proof of 5.6, one has used the fact that a nonempty reduced scheme of finite type over an algebraically closed field admits at least one regular point (hence smooth), a fact which is usually proved by the differential method (via Zariski's theorem that the set of regular points of X is open). If one wants to avoid a vicious circle, one must prove that if K/k is a separable extension of finite type, and if the $f_i \in K$ are such that the $d_{K/k}f_i$ form a basis of $\Omega_{K/k}^1$, ($1 \leq i \leq n$), then n is the transcendence degree of K over k , i.e. the f_i are algebraically independent. The proof of this fact with the aid of the Mac Lane criterion is well known, cf. Bourbaki, Algèbre, Chap. V par. 9 th. 2: one takes a polynomial $g \in k[t_1, \dots, t_n]$ of minimal degree such that $g(f_1, \dots, f_n) = 0$, one has therefore

$$\sum \frac{dg}{dt_i}(f_1, \dots, f_n)df_i = 0$$

whence (since the df_i form a basis of $\Omega_{K/k}^1$) the fact that the dg/dt_i vanish at $(f_1, \dots, f_n)$, hence are zero by the minimal character of g , hence if k is of characteristic 0 one has $g = 0$, and if k is of characteristic $p \neq 0$ one has $g = h(t_1^p, \dots, t_n^p)$. Using the Mac Lane criterion, one sees that the polynomial $h \in k[t_1, \dots, t_n]$ also vanishes at $(f_1, \dots, f_n)$, whence again $g = 0$ by the minimal character of g .